
YI BA

Wuhan University of Technology

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EDUCATION

Wuhan University of Technology

B.Eng. in Vehicle Engineering

GPA: 4.253/5.0 | Major Rank: 2/149 (Top 1.3%)

Wuhan, China

Sep 2024 – Expected Jun 2028

RESEARCH INTERESTS

Robot Learning; Contact-Rich Manipulation; Autonomous Navigation; Shared Control

SUBMITTED MANUSCRIPT

Surface-Corridor-Based Intent-Aware Shared Control for Human-Guided Contact Motion on Curved Surfaces

Yi Ba, Chunhan Wang, Biwei Tang, Muye Pang, Jin Shi, and Jing Luo.

Submitted to *IEEE Transactions on Automation Science and Engineering (T-ASE)*, 2026.

RESEARCH EXPERIENCE

Wuhan University of Technology

Wuhan, China

Undergraduate Researcher, supervised by Prof. Jing Luo

Apr 2026 – Present

CoSurf: Shared Control for Curved-Surface Contact Manipulation

- Designed a trajectory-conditioned surface-corridor frame field from registered point clouds to provide geometry-consistent position and orientation references for off-centerline contact motion.
- Developed confidence-weighted intent arbitration with a haptic device and integrated virtual fixtures (VF) with disturbance-observer (DOB)-enhanced normal admittance control on a Sawyer robot.
- Validated the integrated framework through orientation-reference comparison, DOB and VF ablation, and human-in-the-loop shared-control experiments.

Shanghai Jiao Tong University

Shanghai, China

Research Intern, supervised by Prof. Tong Qin

Jul 2026 – Present

DynaSCAN: Dynamic Collision-Aware Local Planning for Quadruped Robots in Crowded Environments

- Extended SCAN-Planner for dynamic indoor environments by integrating obstacle tracking and motion prediction with B-spline trajectory optimization for the Unitree Go2.
- Designed clearance-aware costs and multi-hypothesis trajectory initialization, with collision validation accounting for robot body geometry and turning dynamics.
- Implemented and evaluated the complete perception-to-planning pipeline in dynamic-obstacle simulations, with broader system validation ongoing.

SELECTED AWARDS AND HONORS

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| • National First Prize, National Marine Vehicle Design and Production Competition | 2025 |
| • National Second Prize, Global Campus Artificial Intelligence Algorithm Elite Competition (AIC) | 2025 |
| • Provincial First Prize, CIMC "Siemens Cup" China Intelligent Manufacturing Challenge | 2025 |

TECHNICAL SKILLS

Programming: Python, C++

Robotics: ROS 1/2, MoveIt, point-cloud processing

Robot Learning: ACT, PPO

Simulation: Gazebo, MuJoCo, Isaac Gym

CAD: SolidWorks, CATIA

Tools: Git, Linux